# Vector-Space Optimization for Contraction Theory-Based Control Design: An Energy-Based Effective Space Approach ⋆

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**Abstract:** In this paper, we propose a computationally efficient and numerically robust contraction metric calculation method for contraction theory-based control design. Conventional Linear Matrix Inequality (LMI) optimization frameworks for contraction metric calculation suffer from high computational complexity for real-time implementation and scale-invariance issues of the objective function which minimizes the condition number of the contraction metric. To address these issues, we project the contraction metrics onto the trajectory error vector space, leading to a simplified vector-space optimization framework within a lower-dimensional effective space. Furthermore, energy-based constraints are incorporated to obtain a unique solution and to maintain sufficient control effort. In addition, convex input saturation constraints are integrated to handle practical limitations of the actuators. The effectiveness and the feasibility of the proposed method are validated through numerical simulations using the Lorenz system.

*Keywords:* Contraction theory, input saturation, energy-based control

## 1. INTRODUCTION

### 1.1 Motivation

Besides the traditional Lyapunov stability theory, contraction theory offers a interesting perspective on the stability of nonlinear systems. It allows us analyze the system stability by examining the convergence of trajectories, not the convergence to a specific equilibrium point Lohmiller and Slotine (1998). Since the contraction property is defined in an differential sense, the Linear Time-Varying (LTV) system analysis techniques can be applied to the contraction theory-based design.

Among recent developments in contraction theory-based control, Tsukamoto et al. proposed ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) to systematically calculate the contraction metrics in Linear Matrix Inequalities (LMIs) optimization problem Tsukamoto and Chung (2021), using the differential nature of contraction theory.

However, since the LMI framework optimizes the contraction metrics in the matrix space, the computational complexity is $\mathcal{O}(n^6)$ [..], which can be a critical issue for high-dimensional systems. Thereby, for the real-time implementation, it is necessary to approximate the contraction metrics using universal function approximators such as neural networks [...].

Moreover, there is no consideration of input saturation in the conventional contraction theory-based control design which limits the practical applicability of the method.

To the best of our knowledge, there has been no previous work that projects the contraction constraints onto a vector space to reduce the computational complexity of the contraction metric calculation. Moreover, input saturation has not been considered in the conventional contraction theory-based control design. Hence, the main objective of this paper is to simplify the contraction metric calculation from matrix-space optimization to vector-space optimization, while also incorporating the input saturation constraint.

### 1.2 Contributions

The main contributions of this paper are

- The contraction metrics can be calculated in a lower-dimensional effective space by projecting the contraction constraints onto the trajectory error vector space, which significantly reduces the computational complexity of the LMI framework for contraction metric calculation, allowing real-time implementation;
- Scale-invariance issues of the objective function which minimizes the condition number of the contraction metric are resolved by introducing energy-based constraints ensures minimum required control effort;
- Convex input saturation constraint is incorporated into the proposed optimization framework, which handles practical limitations of the actuators; and

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- Robust feasibility of the proposed method is enhanced by introducing slack variables to relax the energy-based constraints to avoid infeasibility by input saturation constraint.

### *1.3 Paper Organization*

The remainder of this paper is organized as follows. Section 2 reviews the preliminaries of contraction theory. Section 3 formulates the problem of contraction theory-based control design with input saturation. Section 4 discusses the conventional LMI framework for contraction metric calculation and its issues. Section 5 presents the proposed vector-space optimization framework for contraction metric calculation and provides an illustrative example. Section 6 validates the proposed method through numerical simulations. Finally, Section 7 concludes the paper.

## 2. PRELIMINARIES

Before presenting the proposed method, we first review the contraction theory and related theorem. The detailed explanation of contraction theory can be found in Lohmiller and Slotine (1998); Tsukamoto et al. (2021).

### *2.1 Contraction Property and Contraction Metric*

Consider that we have the following nonlinear system:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t), \tag{1}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ denotes the state variables and $\boldsymbol{f} := \boldsymbol{f}(\boldsymbol{x}, t)$ denotes the nonlinear system function. Additionally, consider differential dynamics of the system (1) which is represented as $\frac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{x} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\delta\boldsymbol{x}$ , (Kirk, 2004, Chap 4).

According to [..], the system (1) is said to be *contracting*, i.e., all trajectories incrementally exponentially converge to a unique trajectory, if the following condition holds:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}\left(\boldsymbol{M}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right) \preceq -2\alpha\boldsymbol{M}, \tag{2}$$

where $\boldsymbol{M}$ denotes *contraction metric* and $\alpha \in \mathbb{R}_{>0}$ denotes the *contraction rate*, and $\mathrm{sym}(\boldsymbol{A}) := \frac{1}{2}(\boldsymbol{A} + \boldsymbol{A}^\top)$. The contraction metric $\boldsymbol{M}$ is symmetric positive definite matrix and satisfies $\overline{m}\boldsymbol{I}_n \succeq \boldsymbol{M} \succeq \underline{m}\boldsymbol{I}_n$ with $\overline{m}, \underline{m} \in \mathbb{R}_{>0}$.

It is notable that the initial conditions are exponentially forgotten in the contracting system Lohmiller and Slotine (1998).

### *2.2 Perturbed Contracting System*

If the system (1) is subjected to bounded disturbances, the system can be expressed as follows:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t) + \boldsymbol{d}(t), \tag{3}$$

where $\boldsymbol{d}(t) \in \mathbb{R}^n$ denotes the external disturbances with $\|\boldsymbol{d}(t)\| \leq \overline{d}$ where $\overline{d} \in \mathbb{R}_{>0}$.

The following theorem states that all solutions of the perturbed system (3) exponentially converge to a bounded ball of the solution of the nominal system (1).

*Theorem 1.* If the system (1) is contracting with the contraction metric $\boldsymbol{M}$ which satisfies the condition (2), the path integral $V_l(\boldsymbol{q}, \delta\boldsymbol{q}, t) = \int_{\boldsymbol{\xi}_1}^{\boldsymbol{\xi}_2}\|\boldsymbol{\Theta}(\boldsymbol{q}, t)\delta\boldsymbol{q}\|$, where $\boldsymbol{\Theta}(\boldsymbol{q}, t) := \boldsymbol{M}^{1/2}$, and $\boldsymbol{\xi}_1$ and $\boldsymbol{\xi}_2$ are the solutions of (1) and (3), respectively, and $\boldsymbol{q}$ is the virtual state variable, exponentially converges to a bounded ball. Especially, the following inequality of a bounded error ball of the solutions of (1) and (3) holds:

$$\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| \leq \tfrac{V_l(0)}{\sqrt{\underline{m}}}\exp(-\alpha t) + \tfrac{\overline{d}}{\alpha}\sqrt{\chi}(1 - \exp(-\alpha t)), \tag{4}$$

where $\chi := \overline{m}/\underline{m}$ denotes the condition number of $\boldsymbol{M}$, yielding the steady state error bound of $\frac{\overline{d}}{\alpha}\chi$.

**Proof.** See the proof of Theorem 2.4 in Tsukamoto et al. (2021).

## 3. PROBLEM FORMULATION

Let us consider the following nonlinear system as follows:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t) + \boldsymbol{B}(\boldsymbol{x}, t)\,\mathrm{sat}(\boldsymbol{u}) + \boldsymbol{d}(t), \tag{5}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{u} \in \mathbb{R}^m$ denote the state variables and the control inputs, respectively, and $\boldsymbol{d} := \boldsymbol{d}(t) \in \mathbb{R}^n$ denotes the bounded external disturbances, i.e., $\|\boldsymbol{d}\| \leq \overline{d}$, and $\boldsymbol{f}(\boldsymbol{x}, t)$ and $\boldsymbol{B}(\boldsymbol{x}, t)$ denote system functions. The input saturation function is denoted by $\mathrm{sat}(\cdot)$ and can be defined as any convex function which is physically feasible.

We assume that some desired control input $\boldsymbol{u}_d \in \mathbb{R}^m$ is pre-designed to achieve the desired trajectory $\boldsymbol{x}_d \in \mathbb{R}^n$ under ideal conditions, i.e., $\boldsymbol{d} = 0$, by simulating the following desired system:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_d = \boldsymbol{f}(\boldsymbol{x}_d, t) + \boldsymbol{B}(\boldsymbol{x}_d, t)\boldsymbol{u}_d. \tag{6}$$

The error dynamics between the system (5) and the desired system (6) can be expressed as follows:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{e} = \boldsymbol{f} - \boldsymbol{f}_d + \boldsymbol{B}\,\mathrm{sat}(\boldsymbol{u}) - \boldsymbol{B}_d\boldsymbol{u}_d + \boldsymbol{d}, \tag{7}$$

where $\boldsymbol{e} := \boldsymbol{x} - \boldsymbol{x}_d$ denotes the trajectory error vector, and the arguments of $\boldsymbol{f}$, $\boldsymbol{f}_d$, $\boldsymbol{B}$, and $\boldsymbol{B}_d$ are omitted for brevity. Using the state-dependent coefficient (SDC) formulation Dani et al. (2015), i.e., $\mathbb{A} := \mathbb{A}(\boldsymbol{x}, \boldsymbol{x}_d) = \int_0^1 \left[\frac{\partial \overline{\boldsymbol{f}}}{\partial \boldsymbol{x}}\right]\big(c\boldsymbol{x} + (1-c)\boldsymbol{x}_d\big)\,\mathrm{d}c$, where $\overline{f} := \boldsymbol{f} - \boldsymbol{f}_d + (\boldsymbol{B} - \boldsymbol{B}_d)\boldsymbol{u}_d$, the error dynamics (7) can be linearized as follows:

$$\begin{aligned}\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{e} =& \underbrace{\boldsymbol{f} - \boldsymbol{f}_d + (\boldsymbol{B} - \boldsymbol{B}_d)\boldsymbol{u}_d}_{=\mathbb{A}(\boldsymbol{x}, \boldsymbol{x}_d)\boldsymbol{e}} + \boldsymbol{B}(\boldsymbol{u} - \boldsymbol{u}_d) + \boldsymbol{B}\boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d} \\ =& \mathbb{A}\boldsymbol{e} + \boldsymbol{B}(\boldsymbol{u} - \boldsymbol{u}_d) + \boldsymbol{B}\boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d},\end{aligned} \tag{8}$$

where $\boldsymbol{\phi}(\boldsymbol{u}) := \mathrm{sat}(\boldsymbol{u}) - \boldsymbol{u}$ denotes input saturation effect.

To simplify the control design, we use the following state feedback control law:

$$\boldsymbol{u} = \boldsymbol{u}_d - \boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M}\boldsymbol{e}, \tag{9}$$

where $\boldsymbol{R} \in \mathbb{R}^{m\times m}$ denotes a positive definite weighting matrix for the control input. Substituting the control law (9) into the error dynamics (8), we obtain:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{e} = \left(\mathbb{A} - \boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M}\right)\boldsymbol{e} + \boldsymbol{B}\boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d}. \tag{10}$$

Finally, the control objective is to obtain the contraction metric $\boldsymbol{M}$ in (9) which guarantees the system (5) to be contracting with respect to the desired trajectory $\boldsymbol{x}_d$ in the presence of input saturation and external disturbances.

## 4. CONVENTIONAL LMI-APPROACH

In Tsukamoto and Chung (2019, 2021), Tsukamoto et al. proposed ConVex optimization-based Steady-state Tracking Error Minimization (CV-STEM) to obtain the contraction metric $\boldsymbol{M}$ in the control input (9) by formulating the LMI optimization problem.

To apply Theorem 1, the author defined the systems (1) and (3) as (5) with $\boldsymbol{d} = \boldsymbol{0}$ and (5), respectively, and the virtual $\boldsymbol{q}$-system as $\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{q} - \boldsymbol{x}_d) = (\mathbb{A} - \boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M})(\boldsymbol{q} - \boldsymbol{x}_d) + \mu\boldsymbol{d}$, $\mu \in [0, 1]$, whose solutions are denoted by $\boldsymbol{q}(\mu = 0, t) =: \boldsymbol{\xi}_1$ and $\boldsymbol{q}(\mu = 1, t) =: \boldsymbol{\xi}_2$, respectively. Then, neglecting input saturation effect, the steady-state error bound between the system (5) and the desired system (6) can be minimized by minimizing the condition number $\chi$ of the contraction metric $\boldsymbol{M}$ proportionally as $\lim_{t\to\infty}\|\boldsymbol{e}(t)\| = \overline{d}\frac{\sqrt{\chi}}{\alpha}$. The contraction constraint (2) in Theorem 1 which should be satisfied by the contraction metric $\boldsymbol{M}$ can be expressed for the error dynamics (10) as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}\left(\boldsymbol{M}\mathbb{A}\right) - 2\boldsymbol{M}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M} \preceq -2\alpha\boldsymbol{M}. \quad (11)$$

Using the variable transformation $\boldsymbol{M} =: \boldsymbol{W}^{-1}$, the following LMI optimization problem can be formulated:

$$\min_{\overline{\boldsymbol{W}},\chi,\nu} \chi + \lambda\nu$$
$$\text{s.t.} \begin{cases} -\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{W}} + 2\,\mathrm{sym}(\mathbb{A}\overline{\boldsymbol{W}}) - 2\nu\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top \preceq -2\alpha\overline{\boldsymbol{W}}, \\ \boldsymbol{I}_n \preceq \overline{\boldsymbol{W}} \preceq \chi\boldsymbol{I}_n, \end{cases} \quad (12)$$

where $\overline{\boldsymbol{W}} := \nu\boldsymbol{W}$ and $\nu \in \mathbb{R}_{>0}$ is a penalty term of 2-norm of $\boldsymbol{M}$. Using $\lambda \in \mathbb{R}_{>0}$, the magnitude of $\boldsymbol{M}$ can be adjusted by $\nu$. Since the LMI optimization problem (12) is convex, the global optimal solution can be obtained.

However, since CV-STEM penalizes the magnitude of the control input indirectly by penalizing $\boldsymbol{M}$, the input saturation constraint is not explicitly considered in (12). In Tsukamoto and Chung (2019), the input saturation constraint is considered in conservative manner by limiting the maximum norm of the control input, not $\boldsymbol{M}$ directly, i.e., $\|\boldsymbol{u}\| \le \overline{u}$ where $\overline{u} \in \mathbb{R}_{>0}$. In other words, Theorem 1 cannot be directly applied to the system (5) with input saturation, since $\boldsymbol{\phi}(\boldsymbol{u})$ is not considered as bounded disturbances.

In the perspective of computational complexity, the computational complexity of solving the LMI optimization problem (12) is quadratically dependent on the dimension. In addition, the LMI framework limits flexibility in handling various constraints including nonlinear input and state constraints. Thus, these issues should be resolved for practical real-time implementation.

In conclusion, though CV-STEM provides a systematic framework for contraction metric calculation in LMI framework, there still remain several issues to be resolved for practical real-time implementation.

*Remark 2.* There exist a severe problem formulation issue in the conventional LMI framework. In the formulation, $\frac{\mathrm{d}}{\mathrm{d}t}\overline{\boldsymbol{W}} = \frac{\mathrm{d}}{\mathrm{d}t}(\nu\boldsymbol{W})$ is considered as same as $\nu\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{W})$, which is not mathematically valid since $\nu$ is also a decision variable. Thus, the conventional LMI framework's solutions may decide $\overline{W}$ and $\chi$ to $\boldsymbol{I}_n$ and 1, respectively, and modifies $\nu$ only.

## 5. MAIN RESULTS

### 5.1 Vector-Space Optimization Framework

In the control law (9), since the contraction metric $\boldsymbol{M}$ is only applied to $\boldsymbol{e}$, we can project the contraction constraints (11) onto the trajectory error vector space. By pre- and post-multiplying $\boldsymbol{e}^\top$ and $\boldsymbol{e}$ to the contraction constraint (11), respectively, we have:

$$\boldsymbol{y}^\top\left(-\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\right)\boldsymbol{y} + \boldsymbol{e}^\top\left(\tfrac{1}{T_s} + 2\alpha + 2\mathbb{A}^\top\right)\boldsymbol{y} - \tfrac{\boldsymbol{e}^\top\boldsymbol{M}_{\mathrm{pre}}\boldsymbol{e}}{T_s} \le 0, \quad (13)$$

where $\boldsymbol{y} := \boldsymbol{M}\boldsymbol{e}$ denotes a new optimization variable. For discrete-time implementation, we approximate $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M}$ as $\frac{1}{T_s}(\boldsymbol{M} - \boldsymbol{M}_{\mathrm{pre}})$ where $T_s$ denotes the sampling time and $\boldsymbol{M}_{\mathrm{pre}}$ denotes the contraction metric obtained at the previous time step.

For input saturation constraint, we define convex constraint function $c(\boldsymbol{y}) = \mathrm{sat}(\boldsymbol{u})$. Notably, the convexity of $c(\boldsymbol{y})$ is preserved, since $\boldsymbol{u}$ is linear with respect to $\boldsymbol{y} = \boldsymbol{M}\boldsymbol{e}$ in the control law (9). Thus, the input saturation constraint can be expressed as a convex constraint function of $\boldsymbol{y}$ as follows:

$$c(\boldsymbol{y}) = \mathrm{sat}(\boldsymbol{u}) \le 0. \quad (14)$$

According to Theorem 1, the steady-state error bound can be proportionally minimized by the condition number of the contraction metric $\boldsymbol{M}$. Alternatively, we define the objective function to minimize the angle between $\boldsymbol{y}$ and $\boldsymbol{e}$, since the optimal condition number $\chi = 1$ is achieved when $\boldsymbol{y}$ is parallel to $\boldsymbol{e}$, resulting $\boldsymbol{M} = \gamma\boldsymbol{I}_n$ for some $\gamma \in \mathbb{R}_{>0}$.

However, there is infinite number of solutions $\boldsymbol{y}$ which is parallel to $\boldsymbol{e}$, i.e., since the condition number is scale-invariant, leading to ill-posed optimization problem. To this end, energy-based constraint is introduced to obtain locally optimal solution, maintaining sufficient control effort as follows:

$$\left(\boldsymbol{e}^\top\boldsymbol{y} - \mu\boldsymbol{e}^\top\boldsymbol{e}\right)^2 \le (\epsilon + s)\|\boldsymbol{e}\|^2, \quad (15)$$

where $\mu \in \mathbb{R}_{>0}$ denotes the desired energy level, and $\epsilon \in \mathbb{R}_{>0}$ and $s \in \mathbb{R}_{\ge 0}$ denote allowable small energy variation and a slack variable to enhance the robust feasibility of the optimization problem.

Thus, the following vector-space optimization problem can be formulated as follows:

$$\min_{\boldsymbol{y},s}\|\boldsymbol{y}\|\|\boldsymbol{e}\| - \boldsymbol{y}^\top\boldsymbol{e} + \lambda s$$
$$\text{s.t. } (13), (14), (15), \quad (16)$$

After the optimization, the contraction metric $\boldsymbol{M}$ can be retrieved maintaining the condition number and the relationship $\boldsymbol{y} = \boldsymbol{M}\boldsymbol{e}$ as follows:

$$\boldsymbol{M} = \tfrac{1}{\sqrt{\chi}}\,\mathrm{diag}\left(\left(\sqrt{\chi}, 1, \ldots, 1, \tfrac{1}{\sqrt{\chi}}\right)\right) \cdot \tfrac{1}{\|\boldsymbol{e}\|\|\boldsymbol{y}\|}, \quad (17)$$

where $\chi = \frac{1+\sin(\theta)}{1-\sin(\theta)}$ denotes the minimum condition number of $\boldsymbol{M}$ when the angle $\theta$ between $\boldsymbol{y}$ and $\boldsymbol{e}$ is given.

The overall procedure of the proposed method is summarized in Algorithm 1 in which the optimization problem (16) is solved at each time step.

**Algorithm 1** Proposed controller

**Initialize:** $\boldsymbol{y}_0 = \mathbf{1}_{n\times 1}$, $s_0 = 0$, $\boldsymbol{M}_0 = \mu \boldsymbol{I}_n$
**for** $k = 1$ to $\infty$ **do**
  Get current $\boldsymbol{x}_k$, $\boldsymbol{x}_{d,k}$, $\boldsymbol{u}_{d,k}$ and $\boldsymbol{e}_k$
  Assign $\boldsymbol{y}_k \leftarrow \boldsymbol{y}_{k-1}$, $s_k \leftarrow s_{k-1}$ for warm-start
  Solve (16) to get optimal $\boldsymbol{y}_k$ and $s_k$
  Calculate $\boldsymbol{M}_k$ using $\boldsymbol{y}_k$ and $\boldsymbol{e}_k$ using (17)
  Apply the control law (9) using $\boldsymbol{M}_k$
**end for**

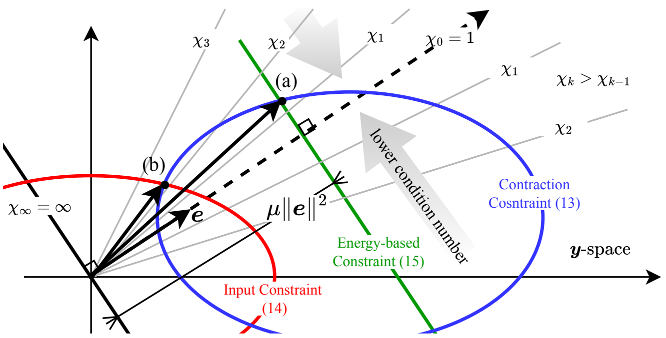


Fig. 1. Illustration of the optimization problem (16) in two-dimensional space, see Section 5.2 for details.

### *5.2 Illustrative Example*

To facilitate the geometric interpretation, the optimization problem (16) is illustrated in two-dimensional space in Fig. 1. The contraction constraint (13) and the input saturation constraint (14) restrict the feasible space of $\boldsymbol{y}$ in the outside of blue line and the inside of red line, respectively. The green line represents the energy-based constraint (15) assuming that the allowable energy variation $\epsilon$ is approximately zero.

Since contraction constraint (13) is ellipse-shaped concave function and input saturation constraint (14) is convex function, there exist at most two intersection points of these constraints. If the initial guess is set near point (a), the optimal solution converges to point (a), neglecting the input saturation constraint according to the objective function, i.e., minimum angle between $\boldsymbol{y}$ and $\boldsymbol{e}$. Considering the input saturation constraint, the optimal solution is obtained at the point (b) by increasing the slack variable $s$, which relaxes the energy-based constraint (15) to satisfy input saturation constraint.

## 6. NUMERICAL VALIDATION

### *6.1 Numerical Validation Set Up*

We employed the Lorenz system to validate the proposed method which is represented as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=:\boldsymbol{x}} = \underbrace{\begin{pmatrix} \sigma(y-x) \\ x(\rho - z) - y \\ xy - \beta z \end{pmatrix}}_{=:\boldsymbol{f}(\boldsymbol{x})} + \mathrm{sat}\left(\underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=:\boldsymbol{u}}\right) + \boldsymbol{d}(t), \quad (18)$$

where $\sigma = 10$, $\rho = 28$, and $\beta = \frac{8}{3}$ are system parameters. The input saturation function $\mathrm{sat}(\cdot)$ was defined as follows:

$$\mathrm{sat}(\boldsymbol{u}) := \begin{cases} \boldsymbol{u}, & \text{if } \|\boldsymbol{u}\| \le \overline{u}, \\ \overline{u}\frac{\boldsymbol{u}}{\|\boldsymbol{u}\|}, & \text{otherwise,} \end{cases} \quad (19)$$

Table 1. Controllers for comparative study.

| | Description |
|---|---|
| ($C_1$) [—] | Proposed controller. |
| ($C_2$) [—] | CV-STEM (Tsukamoto and Chung (2021)). |
| ($C_3$) [—] | Desired controller, i.e., $\boldsymbol{u} = \boldsymbol{u}_d$. |

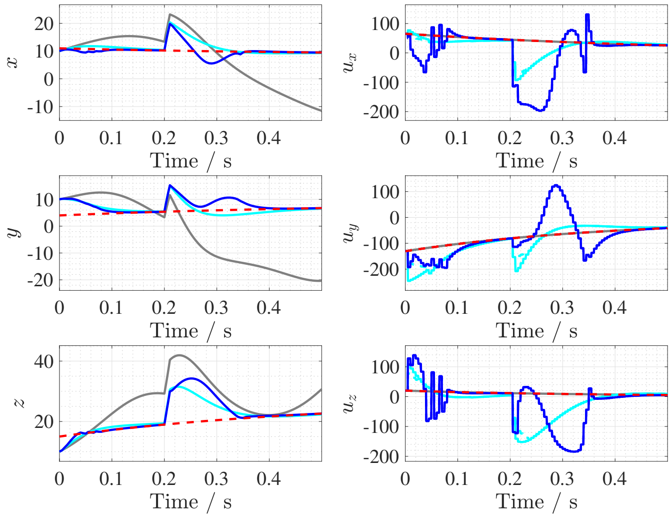


Fig. 2. Numerical validation results of ($C_1$) [—], ($C_2$) [—], ($C_3$) [—], and desired trajectory $x_d$ [- - -]

where $\overline{u} = 200$ denotes the maximum input magnitude. The external disturbances $\boldsymbol{d}(t)$ were modeled as follows:

$$\boldsymbol{d}(t) := 1000(H(t-0.1) - H(t-0.11)) \cdot \mathbf{1}_{3\times 1}, \quad (20)$$

where $H(\cdot)$ denotes the Heaviside step function.

For the desired trajectory, we simulated the desired system (6) with the desired feedback-linearization control input $\boldsymbol{u}_d := -\boldsymbol{f}(\boldsymbol{x}_d) - 2(\boldsymbol{x}_d - \boldsymbol{r})$, where $\boldsymbol{r} := (-8.485, 8.485, 27)^\top$ denotes the reference point. The desired input was designed not to exceed the input saturation limit.

The proposed method ($C_1$) was compared with CV-STEM ($C_2$) Tsukamoto and Chung (2021) and the desired controller, i.e., $\boldsymbol{u} = \boldsymbol{u}_d$, ($C_3$) as summarized in Table 1. The parameters shared by ($C_1$) and ($C_2$) were $\alpha = 10$ and $\boldsymbol{R} = \boldsymbol{I}_3$. In addition, parameters were selected as $\mu = 30$ and $\lambda = 100$ for ($C_1$) and $\lambda = 0.01$ for ($C_2$). To solve the optimization problems, `fmincon` function and `SeMeDi` toolbox were used for ($C_1$) and ($C_2$) in MATLAB R2024b, respectively.

The simulation was conducted for $T = 0.5$ seconds with the sampling time of $T_s = 0.005$ seconds for the controllers and 10 times finer time step for the system simulation. The initial conditions were set to $\boldsymbol{x}(0) = (10, 5, 13)^\top$ and $\boldsymbol{x}_d(0) = (11, 4, 15)^\top$.

To facilitate reproducibility, the numerical validation source code is publicly available at `https://github.com/KAIST-MIC-Lab/Deep-Neuro-Control`

### *6.2 Numerical Validation Results*

The numerical validation results are illustrated in Figs. 2 and 3. When only the desired controller was applied, i.e., ($C_3$), the trajectories of ($C_3$) and $\boldsymbol{x}_d$ does not coincide due

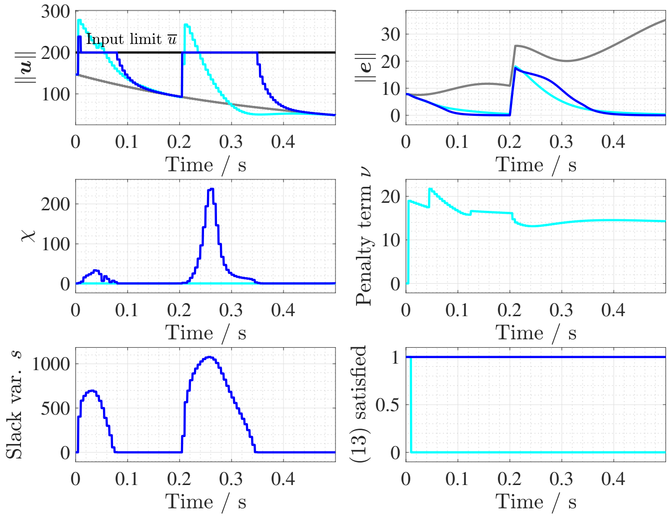


Fig. 3. Control input norms, condition numbers, penalty term and slack variable.

Table 2. Tracking performances in $L_2$ norm.

| | $(C_1)$ [▬] | $(C_2)$ [▬] | $(C_3)$ [▬] |
|---|---|---|---|
| $\int_0^T \|e\|\,\mathrm{d}t$ | 0.4937 (-87.82%) | 0.7929 (-80.44%) | 4.0534 (–) |

to the effect of initial condition mismatch. Significantly, after the disturbance occurrence at $t = 0.2$ seconds, the trajectory deviates further from the desired trajectory as shown in Fig. 2.

By applying the contraction-based controllers, both $(C_1)$ and $(C_2)$ successfully drive the trajectory to closely follow the desired trajectory $\boldsymbol{x}_d$ even in the presence of disturbances, minimizing $L_2$ norm of the tracking error about 87.82% and 80.44% compared to $(C_3)$ as summarized in Table 2. However, in terms of the input saturation, $(C_2)$ reaches the saturation limit by the initial condition mismatch and the disturbance occurs, while $(C_1)$ effectively avoids input saturation as shown in Fig. 3. Moreover, $(C_1)$ effectively tracked the desired trajectory faster than $(C_2)$ while satisfying the input saturation constraint. This demonstrates the effectiveness of the proposed method in handling input saturation constraints while achieving satisfactory tracking performance.

In addition, the expected condition number $\chi$ behaviors of the contraction metrics in Section 5 through Fig. 1 of $(C_1)$ can be observed in Fig. 3. As the contraction constraint (11) and the input saturation constraint (14) become active, the angle between $\boldsymbol{y}$ and $\boldsymbol{e}$ increases, resulting in the increase of the condition number. Specifically, when the input saturation constraint became active, the condition number $\chi$ increased from 1 and decreased again to 1 after the saturation effect disappeared.

In contrast, $(C_2)$ shows a smallest condition number, $\chi = 1$, due to the problem formulation issue discussed in Remark 2. Instead of adjusting the condition number, $(C_2)$ increases the penalty term $\nu$ to increase the control effort as shown in Fig. 3, while making $\overline{W}$ close to identity matrix, i.e., $\boldsymbol{M} = \nu \boldsymbol{I}_n$. This shows the issues of the conventional optimization formulation discussed in Remark 2.

## 7. CONCLUSION

In this study, we presented a vector-space optimization framework for contraction metric calculation in contraction theory-based control design. By projecting the contraction constraints onto the trajectory error vector space, the contraction metrics can be calculated in a lower-dimensional effective space, resolving the scalability issue of the conventional LMI framework. The optimization problem was formulated to minimize the condition number of the contraction metric, where energy-based constraints were introduced to resolve the scale-invariance ambiguity and ensure sufficient control effort. Additionally, convex input saturation constraints were easily incorporated into the proposed framework owing to its flexibility compared to the conventional LMI framework. As a future work, we will extend the proposed method to handle state constraints and validate through real-world experiments.

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